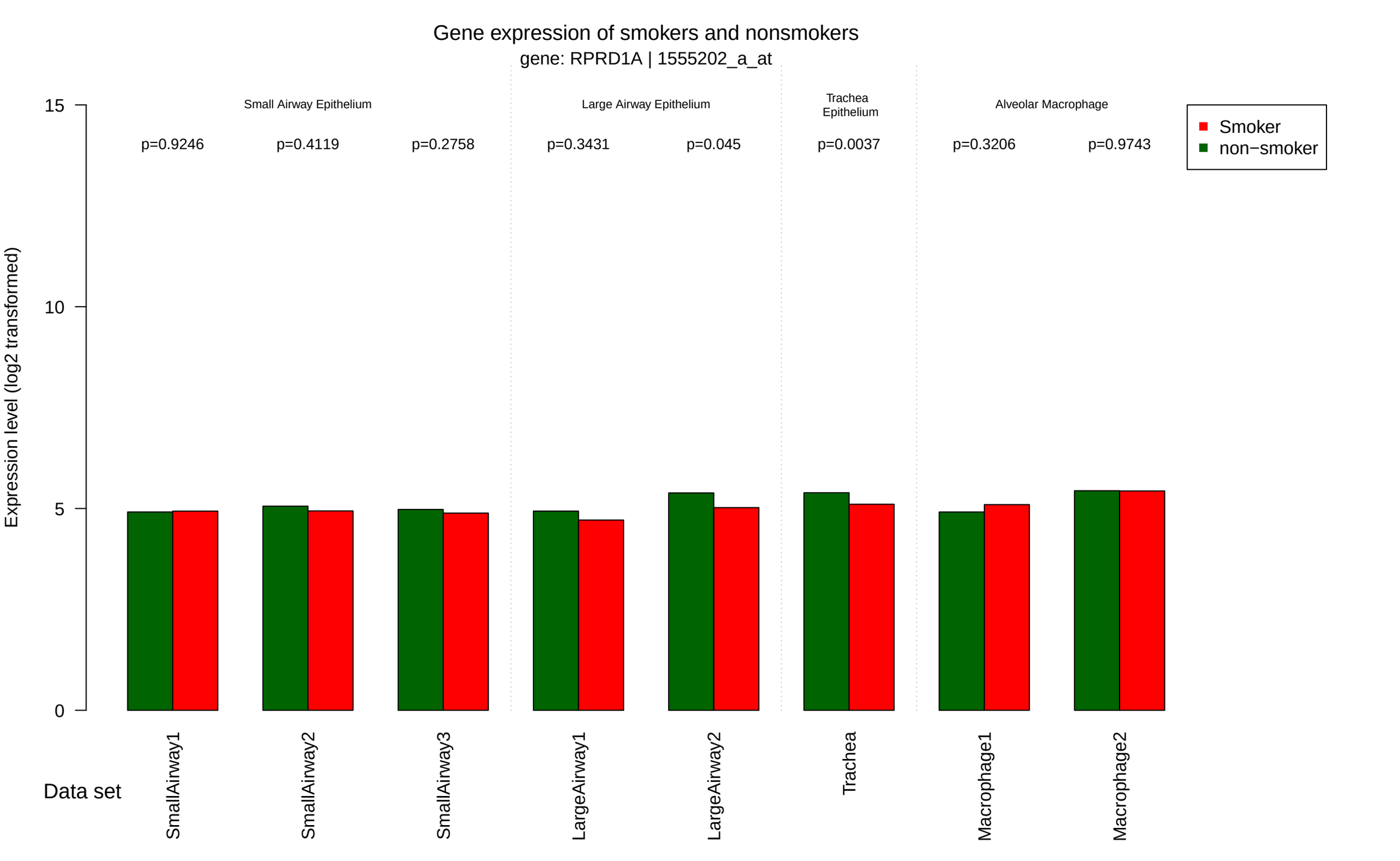
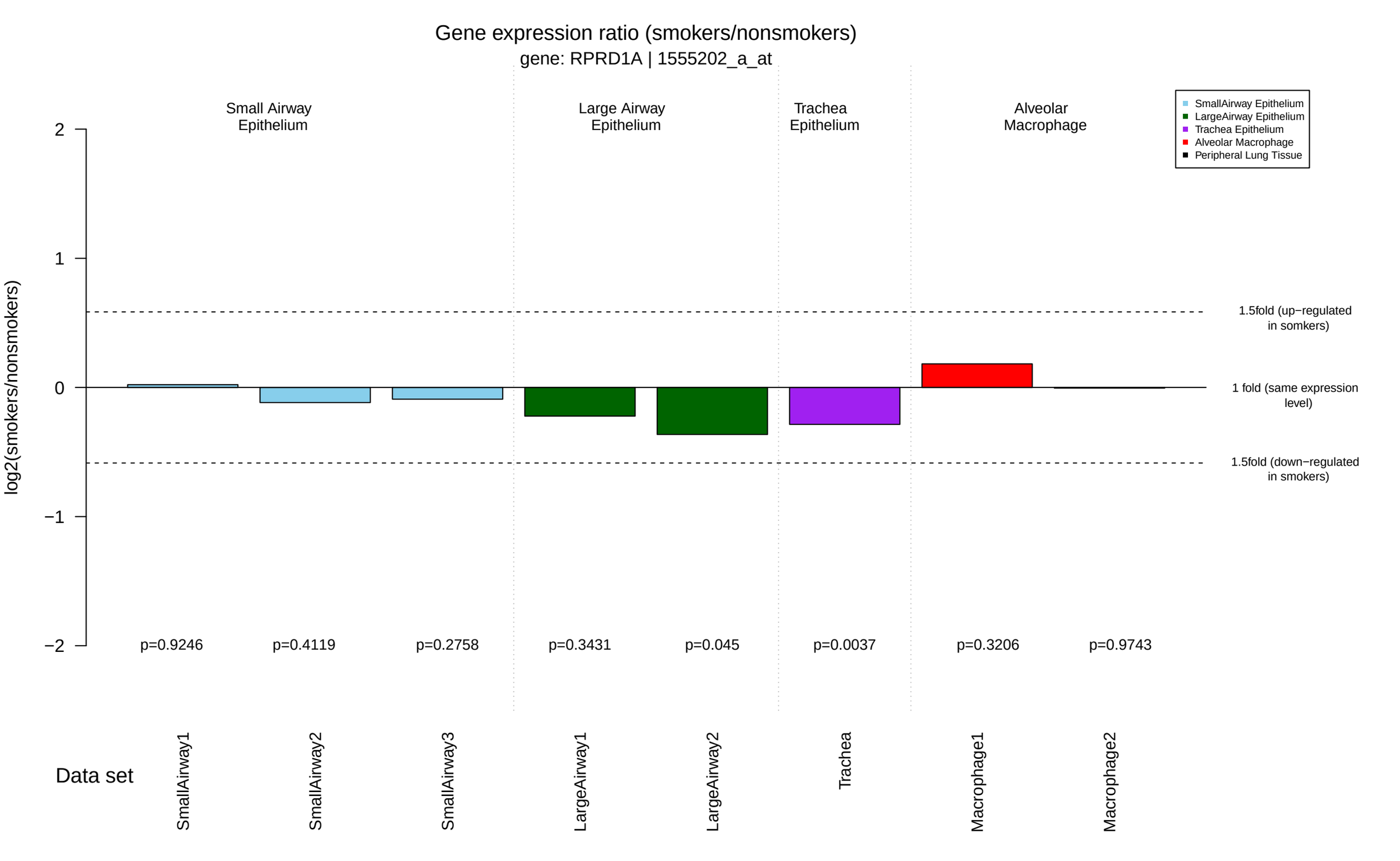
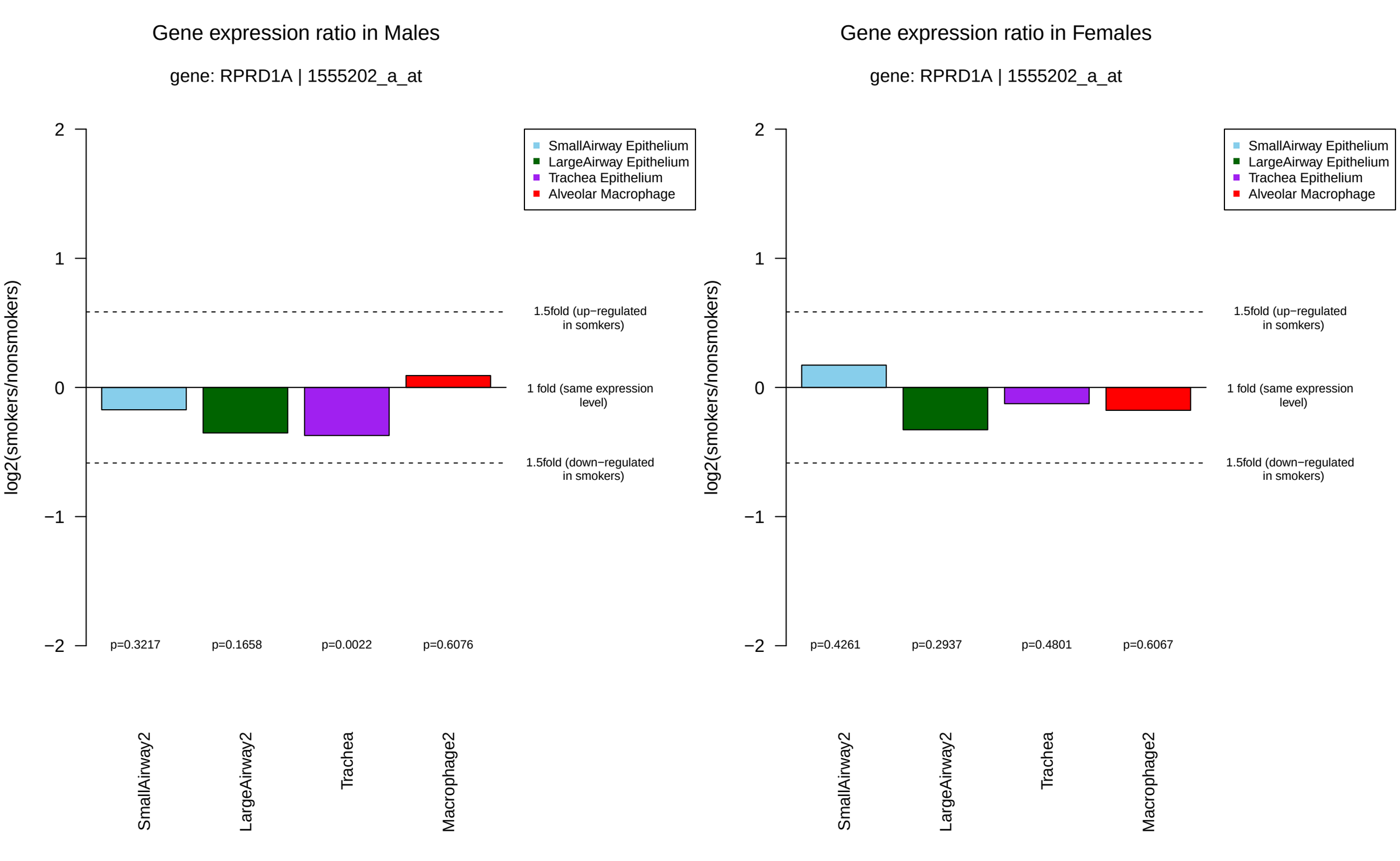
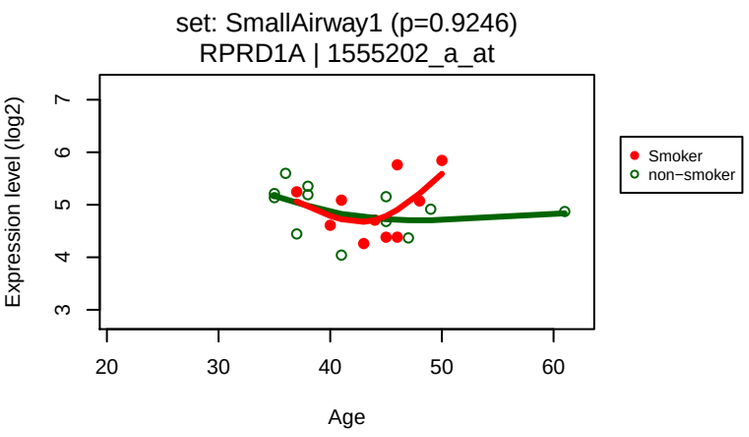
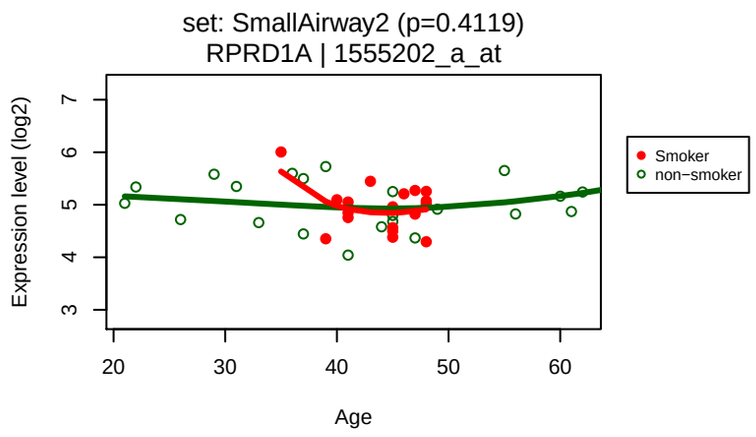
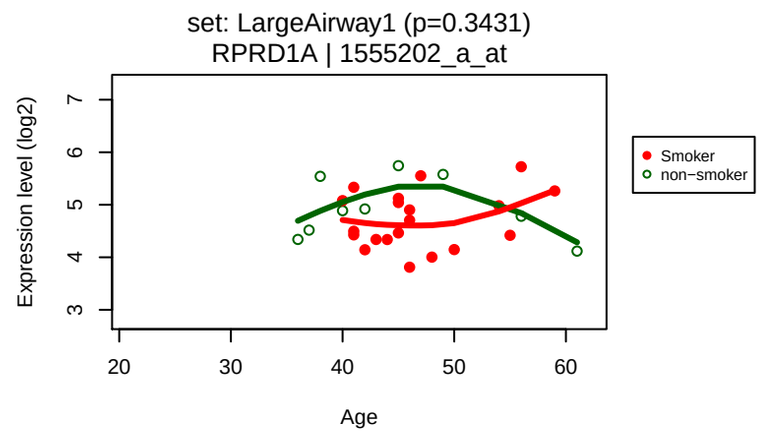

Gene expression of smokers and nonsmokers
gene: RPRD1A | 1555202_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9246
p=0.4119
p=0.2758
p=0.3431
p=0.045
p=0.0037
p=0.3206
p=0.9743
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

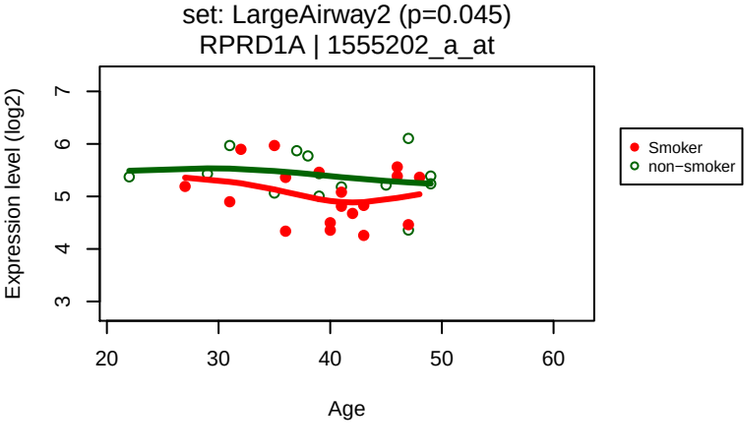

Gene expression ratio (smokers/nonsmokers)
gene: RPRD1A | 1555202_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9246
p=0.4119
p=0.2758
p=0.3431
p=0.045
p=0.0037
p=0.3206
p=0.9743
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

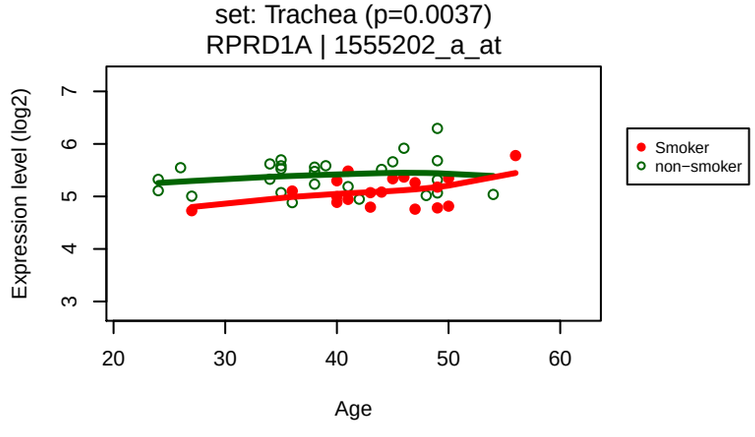

Gene expression ratio in Males
gene: RPRD1A | 1555202_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3217
p=0.1658
p=0.0022
p=0.6076
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD1A | 1555202_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4261
p=0.2937
p=0.4801
p=0.6067
SmallAirway2
LargeAirway2
Trachea
Macrophage2

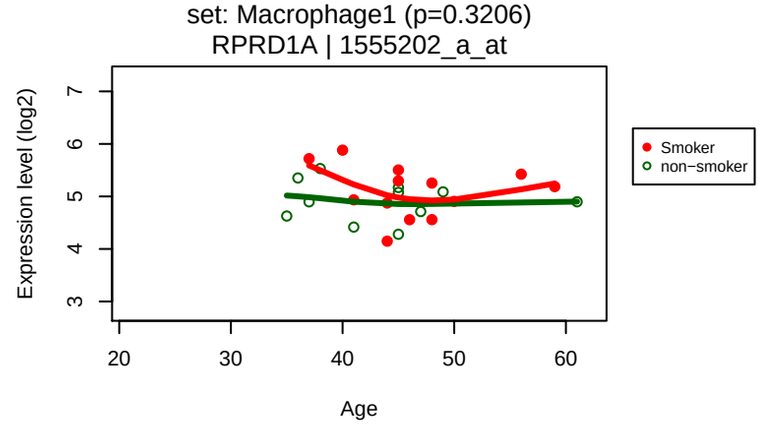

set: SmallAirway1 (p=0.9246)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker

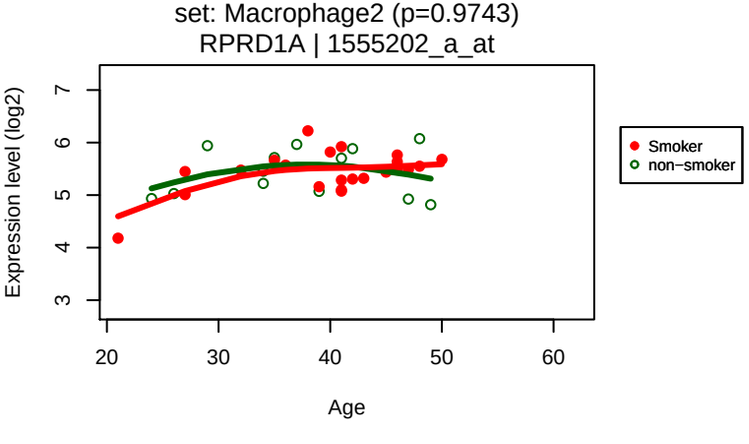

set: SmallAirway2 (p=0.4119)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker

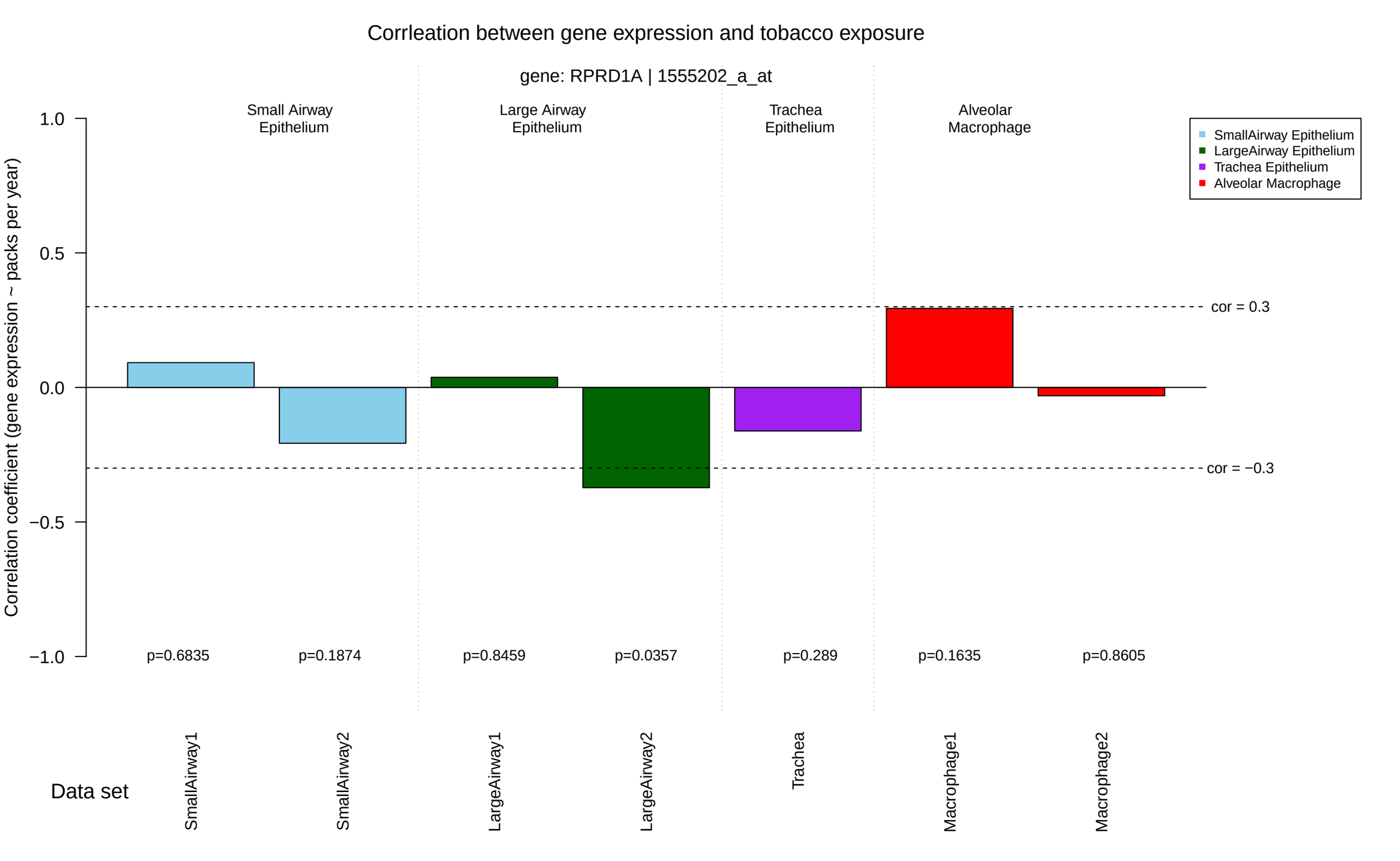

set: LargeAirway1 (p=0.3431)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.045)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3206)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9743)
RPRD1A | 1555202_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD1A | 1555202_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6835
p=0.1874
p=0.8459
p=0.0357
p=0.289
p=0.1635
p=0.8605
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2